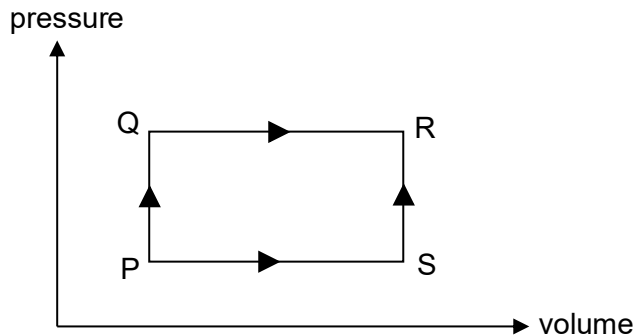


14 A fixed mass of ideal gas undergoes changes of pressure and volume as shown below.



When the gas is taken from state P to state R by the stages PQ and QR, 8 J of heat are absorbed by it and 3 J of work are done by it. When the same resultant change is achieved by stages PS and SR, 1 J of work is done by the gas. In this case,

$$\Delta U = Q_{PQR} + W_{PQR}$$

$$\Delta U = 8 - 3 = 5$$

- A 12 J of heat are liberated.
- B 10 J of heat are absorbed.
- C 8 J of heat are absorbed.
- D 6 J of heat are absorbed.

$$\Delta U = Q_{PSR} + W_{PSR}$$

$$5 = Q_{PSR} - 1$$

$$Q_{PSR} = 6$$

10

Ans: D

15 A beam of light goes through a polarizing filter, then through a sugar solution, then through a